UCC
University College Cork, Ireland
Coláiste na hOllscoile Corcaigh

Workplace Technology and Skills (CS3306)

Conditional Statements and Loops in Shell Scripting

Lecture 10

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The if Statement

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Basic Syntax:

```
if condition
then
    commands
fi
```

With else/elif:

```
if [ -f "myfile.txt" ]
then
    echo "File exists"
elif [ -d "mydir" ]
then
    echo "Directory exists"
else
    echo "Not found"
fi
```

Numeric Comparisons

Operator	Meaning
-eq	Equal to
-ne	Not equal to
-gt	Greater than
-ge	Greater than or equal
-lt	Less than
-le	Less than or equal

```
if [ $age -ge 18 ]
then
    echo "Adult"
fi
```

File Tests

Operator	Meaning
-e	File exists
-f	Regular file exists
-d	Directory exists
-r	File is readable
-w	File is writable
-x	File is executable

```
if [ -x "script.sh" ]
then
    ./script.sh
fi
```

String Comparisons

Operator	Meaning
=	Equal
!=	Not equal
-z	String is empty
-n	String is not empty

```
if [ "$name" = "Alice" ]
then
    echo "Hello, Alice!"
fi
```

Logical Operators

Operator	Alt	Meaning
-a	&&	AND
-o	||	OR
!		NOT

```
# AND operator
if [ $age -ge 18 ] && [ "$name" = "Alice" ]
then
    echo "Alice is an adult"
fi

# NOT operator
if [ ! -f "myfile.txt" ]
then
    touch myfile.txt
fi
```

The while Loop

Executes while condition is true.

```
counter=1
while [ $counter -le 5 ]
do
    echo "Count: $counter"
    ((counter++))
done

# Read file line by line
while read line
do
    echo "Line: $line"
done < input.txt
```

The for Loop

Iterates over a list of items.

```
# Iterate over list
for fruit in apple banana cherry
do
    echo "Fruit: $fruit"
done

# Iterate over files
for file in *.txt
do
    echo "Processing: $file"
done

# Range
for num in {1..5}
do
    echo $num
done
```

The case Statement

Alternative to multiple if-elif statements.

```
read -p "Enter a color: " color

case $color in
    red)
        echo "You chose red"
        ;;
    blue)
        echo "You chose blue"
        ;;
    *)
        echo "Unknown color"
        ;;
esac
```

Additional Resources

- **Bash Guide**: `https://www.gnu.org/software/bash/manual/`
- **Tutorial**: `https://tldp.org/LDP/abs/html/tests.html`
- **Book**: "Classic Shell Scripting" by Robbins